

DSA Sheet 2 - Problems 21 to 40

--- Problem 21: Number of Good Pairs ---

Problem Statement

Given an array of integers `nums`, return the number of good pairs.

A pair (i, j) is called good if:

`nums[i] == nums[j]`

`i < j`

Your task is to count all such good pairs in the array.

Input Format

The first line contains an integer `n` - the size of the array.

The second line contains `n` space-separated integers representing the array `nums`.

Output Format

Print a single integer representing the number of good pairs.

Constraints

`1 <= nums.length <= 100`

`1 <= nums[i] <= 100`

Sample Input

6

1 2 3 1 1 3

Sample Output

4

Explanation

The good pairs are:

(0,3) because `nums[0] = nums[3] = 1`

(0,4) because `nums[0] = nums[4] = 1`

(3,4) because `nums[3] = nums[4] = 1`

(2,5) because `nums[2] = nums[5] = 3`

So, the total number of good pairs is 4.

--- Problem 22: Group Anagrams ---

Problem Statement

Given an array of strings `strs`, group the anagrams together. You can return the answer in any order.

Two strings are called anagrams if they contain the same characters with the same frequencies, but arranged in a different order.

Input Format

The first line contains an integer `n` - the number of strings.

The second line contains `n` space-separated strings representing the array `strs`.

Output Format

Print the grouped anagrams as a list of lists.

Constraints

`1 <= strs.length <= 10^4`

`0 <= strs[i].length <= 100`

`strs[i]` consists of lowercase English letters only.

Sample Input

6

eat tea tan ate nat bat

Sample Output

`[["bat"], ["nat", "tan"], ["ate", "eat", "tea"]]`

DSA Sheet 2 - Problems 21 to 40

Explanation

"bat" cannot be rearranged to form any other word in the list.

"nat" and "tan" are anagrams because both contain the letters a, n, and t.

"ate", "eat", and "tea" are anagrams because they contain the same letters arranged differently.

--- Problem 23: Subsequences with Min + Max $\leq$ Target ---

Problem Statement

You are given an array of integers `nums` and an integer `target`.

Return the number of non-empty subsequences of `nums` such that:

The sum of the minimum element and maximum element in the subsequence is less than or equal to `target`.

Since the answer may be very large, return it modulo $10^9 + 7$.

Input Format

The first line contains an integer `n` - the size of the array.

The second line contains `n` space-separated integers representing the array `nums`.

The third line contains an integer `target`.

Output Format

Print a single integer representing the number of valid subsequences.

Constraints

$1 \leq \text{nums.length} \leq 10^5$

$1 \leq \text{nums}[i] \leq 10^6$

$1 \leq \text{target} \leq 10^9$

Sample Input

```
4
3 5 6 7
9
```

Sample Output

```
4
```

Explanation

The valid subsequences are:

```
[3]
[3,5]
[3,6]
[3,5,6]
```

For each subsequence, the sum of the minimum and maximum elements is less than or equal to 9.

So, the total number of valid subsequences is 4.

--- Problem 24: Container With Most Water ---

Problem Statement

You are given an integer array `height` of length `n`.

There are `n` vertical lines drawn such that the two endpoints of the `i`-th line are:

(`i`, 0)

(`i`, `height[i]`)

Find two lines that together with the x-axis form a container such that the container contains the maximum amount of water.

Return the maximum amount of water a container can store.

Input Format

The first line contains an integer `n` - the size of the array.

The second line contains `n` space-separated integers representing the array `height`.

Output Format

Print a single integer representing the maximum amount of water the container can store.

Constraints

DSA Sheet 2 - Problems 21 to 40

$2 \leq \text{height.length} \leq 10^5$

$0 \leq \text{height}[i] \leq 10^4$

Sample Input

9

1 8 6 2 5 4 8 3 7

Sample Output

49

Explanation

The lines at index 1 and index 8 form the container with the maximum water.

Width = $8 - 1 = 7$

Height = $\min(8, 7) = 7$

So, the maximum water stored is: $7 * 7 = 49$

--- Problem 25: Reverse String ---

Problem Statement

You are given a string s , and your task is to reverse the string.

Input Format

A single line containing the string s .

Output Format

Print the reversed string.

Constraints

$1 \leq s.size() \leq 10^6$

Sample Input

pratik

Sample Output

kitarp

Explanation

The reverse of "pratik" is "kitarp".

--- Problem 26: Palindrome String ---

Problem Statement

You are given a string s . Your task is to determine whether the string is a palindrome.

A string is considered a palindrome if it reads the same forwards and backwards.

Input Format

A single line containing the string s .

Output Format

Print true if the string is a palindrome, otherwise print false.

Constraints

$1 \leq s.size() \leq 10^6$

The string contains only lowercase English letters (a-z).

Sample Input

abba

Sample Output

true

DSA Sheet 2 - Problems 21 to 40

Explanation

"abba" reads the same from both forward and backward directions, so it is a palindrome.

--- Problem 27: Reverse Words in a String ---

Problem Statement

Given an input string *s*, reverse the order of the words.

A word is defined as a sequence of non-space characters. The words in *s* will be separated by at least one space.

Return a string of the words in reverse order concatenated by a single space.

Note:

s may contain leading or trailing spaces.

s may contain multiple spaces between two words.

The returned string should only have a single space separating the words.

Do not include any extra spaces.

Input Format

A single line containing the string *s*.

Output Format

Print the string after reversing the order of the words.

Constraints

$1 \leq s.length \leq 10^4$

s contains English letters (upper-case and lower-case), digits, and spaces ' '.

There is at least one word in *s*.

Sample Input

the sky is blue

Sample Output

blue is sky the

Explanation

The words are reversed and joined using a single space.

--- Problem 28: Rotate String ---

Problem Statement

Given two strings *s* and *goal*, return true if and only if *s* can become *goal* after some number of shifts on *s*.

A shift on *s* consists of moving the leftmost character of *s* to the rightmost position.

For example: If *s* = "abcde", then after one shift it becomes "bcdea".

Input Format

The first line contains the string *s*.

The second line contains the string *goal*.

Output Format

Print true if *s* can be rotated to become *goal*, otherwise print false.

Constraints

$1 \leq s.length, goal.length \leq 100$

s and *goal* consist of lowercase English letters.

Sample Input

abcde

cdeab

Sample Output

true

DSA Sheet 2 - Problems 21 to 40

Explanation

After performing two shifts on "abcde":

First shift -> "bcdea"

Second shift -> "cdeab"

The resulting string matches goal, so the answer is true.

--- Problem 29: First Non-Repeating Character ---

Problem Statement

Given a string *s* consisting of lowercase English letters, return the first non-repeating character in the string.

If there is no non-repeating character, return '\$'.

Input Format

A single line containing the string *s*.

Output Format

Print the first non-repeating character.

If every character repeats, print '\$'.

Constraints

$1 \leq s.size() \leq 10^5$

Sample Input

geeksforgeeks

Sample Output

f

Explanation

In the given string, 'f' is the first character that does not repeat.

--- Problem 30: Roman to Integer ---

Problem Statement

Roman numerals are represented by seven different symbols:

Symbol Value

I	1
V	5
X	10
L	50
C	100
D	500
M	1000

Roman numerals are usually written from largest to smallest from left to right. However, in some cases subtraction is used:

I can be placed before V and X to make 4 and 9.

X can be placed before L and C to make 40 and 90.

C can be placed before D and M to make 400 and 900.

Given a Roman numeral string *s*, convert it into an integer.

Input Format

A single line containing the Roman numeral string *s*.

Output Format

Print the integer value of the Roman numeral.

Constraints

$1 \leq s.length \leq 15$

s contains only the characters ('I', 'V', 'X', 'L', 'C', 'D', 'M')

It is guaranteed that *s* is a valid Roman numeral in the range [1, 3999].

DSA Sheet 2 - Problems 21 to 40

Sample Input

III

Sample Output

3

Explanation

$III = I + I + I = 3$

--- Problem 31: Pangram Check ---

Problem Statement

Check if the given string S is a Pangram or not.

A Pangram is a sentence containing every letter of the English alphabet from a to z at least once.

Return:

1 if the string is a Pangram

0 otherwise

Input Format

A single line containing the string S.

Output Format

Print 1 if the string is a Pangram, otherwise print 0.

Constraints

$1 \leq |S| \leq 10^5$

Sample Input

Pack my box with five dozen liquor jugs

Sample Output

1

Explanation

The given string contains all English alphabets from a to z, so it is a Pangram.

--- Problem 32: Validate IPv4 Address ---

Problem Statement

You are given a string S in the form of an IPv4 Address.

Your task is to validate the IPv4 address:

Return true if it is a valid IPv4 address.

Otherwise, return false.

An IPv4 address is represented in dot-decimal notation and consists of four decimal numbers separated by dots (.).

Each number:

Must be in the range 0 to 255

Must not contain leading zeroes (except the number 0 itself)

A valid IPv4 address follows the format: $x_1.x_2.x_3.x_4$

where each x_i satisfies: $0 \leq x_i \leq 255$

Examples of valid IP addresses: 172.16.254.1, 192.168.1.1

Examples of invalid IP addresses:

256.100.10.1 -> number greater than 255

01.2.3.4 -> leading zero

1.2.3 -> less than 4 parts

Input Format

A single line containing the string S.

DSA Sheet 2 - Problems 21 to 40

Output Format

Print true if the IP address is valid, otherwise print false.

Constraints

$1 \leq |S| \leq 15$

Sample Input

222.111.111.111

Sample Output

true

Explanation

All four parts are present, and every number lies between 0 and 255. Therefore, the given IP address is valid.

--- Problem 33: Add Binary ---

Problem Statement

Given two binary strings a and b, return their sum as a binary string.

Both strings contain only '0' and '1' characters.

Input Format

The first line contains the binary string a.

The second line contains the binary string b.

Output Format

Print the binary sum of a and b.

Constraints

$1 \leq a.length, b.length \leq 10^4$

a and b consist only of '0' or '1' characters.

Each string does not contain leading zeroes except for the number 0 itself.

Sample Input

11

1

Sample Output

100

Explanation

Binary addition:

11

+ 1

100

So, the binary addition results in "100".

--- Problem 34: Look-and-Say Sequence ---

Problem Statement

Given an integer n, return the n-th term of the Look-and-Say sequence.

The sequence starts as:

1

11

21

1211

111221

...

DSA Sheet 2 - Problems 21 to 40

To generate the next term, read the previous term and describe the count of consecutive digits.

For example:

"1" is read as "one 1" -> "11"

"11" is read as "two 1s" -> "21"

"21" is read as "one 2, one 1" -> "1211"

Input Format

A single integer n.

Output Format

Print the n-th term of the Look-and-Say sequence.

Constraints

$1 \leq n \leq 30$

Sample Input

5

Sample Output

111221

Explanation

The sequence generation is:

Step 1: 1

Step 2: 11

Step 3: 21

Step 4: 1211

Step 5: 111221

So, the 5th term of the pattern is "111221".

--- Problem 35: Interleaving Strings ---

Problem Statement

Given strings s1, s2, and s3, find whether s3 is formed by an interleaving of s1 and s2.

An interleaving of two strings s and t is a configuration where they are divided into non-empty substrings such that:

$s = s_1 + s_2 + \dots + s_n$

$t = t_1 + t_2 + \dots + t_m$

and the interleaving is formed as:

$s_1 + t_1 + s_2 + t_2 + \dots$

or

$t_1 + s_1 + t_2 + s_2 + \dots$

Note: a + b represents the concatenation of strings a and b.

Return:

true if s3 is formed by interleaving s1 and s2

false otherwise

Input Format

The first line contains the string s1.

The second line contains the string s2.

The third line contains the string s3.

Output Format

Print true if s3 is an interleaving of s1 and s2, otherwise print false.

Constraints

$0 \leq s1.length, s2.length \leq 100$

$0 \leq s3.length \leq 200$

s1, s2, and s3 consist of lowercase English letters.

DSA Sheet 2 - Problems 21 to 40

Sample Input

aabcc
dbbca
aadbcbcbac

Sample Output

true

Explanation

One possible way to form s3 is:

s1 = "aa" + "bc" + "c"

s2 = "dbbc" + "a"

Interleaving them: "aa" + "dbbc" + "bc" + "a" = "aadbcbcbac"

Hence, s3 can be formed by interleaving s1 and s2.

--- Problem 36: ASCII Value of Character ---

Problem Statement

Given a character, find the ASCII value of the character.

Input Format

A single character C.

Output Format

Print the ASCII value of the given character.

Constraints

The input contains only one valid character.

Sample Input

A

Sample Output

65

Explanation

The ASCII value of character 'A' is 65.

--- Problem 37: Capitalize First and Last of Each Word ---

Problem Statement

Given a string, write a program to capitalize the first and last character of each word in the string.

Input Format

A single line containing the string s.

Output Format

Print the modified string after capitalizing the first and last character of every word.

Constraints

The string contains words separated by spaces.

Sample Input

hello world

Sample Output

HeIlO WoRlD

Explanation

Word 1: "hello" -> First = h, Last = o -> "HeIlO"

DSA Sheet 2 - Problems 21 to 40

Word 2: "world" -> First = w, Last = d -> "World"

--- Problem 38: Sliding Window Maximum ---

Problem Statement

Given an array `arr[]` of positive integers and an integer `k`, find the maximum value for each contiguous subarray of size `k`. Return an array containing the maximum values of all contiguous subarrays.

Input Format

The first line contains an integer `n` - the size of the array.

The second line contains `n` space-separated integers representing the array `arr`.

The third line contains an integer `k`.

Output Format

Print the maximum values of each contiguous subarray of size `k`.

Constraints

$1 \leq \text{arr.size()} \leq 10^6$

$1 \leq k \leq \text{arr.size()}$

$0 \leq \text{arr}[i] \leq 10^9$

Sample Input

```
8
1 2 3 1 4 5 2 3
3
```

Sample Output

```
[3, 3, 4, 5, 5, 5]
```

Explanation

Contiguous subarrays of size 3 and their maximum values:

[1, 2, 3] -> 3

[2, 3, 1] -> 3

[3, 1, 4] -> 4

[1, 4, 5] -> 5

[4, 5, 2] -> 5

[5, 2, 3] -> 5

Therefore, the output is: [3, 3, 4, 5, 5, 5]

--- Problem 39: Divide Two Integers ---

Problem Statement

Given two integers `dividend` and `divisor`, divide the `dividend` by the `divisor` without using multiplication, division, and mod operators. Return the quotient after dividing `dividend` by `divisor`.

Note:

If the quotient is strictly greater than $2^{31} - 1$, return $2^{31} - 1$.

If the quotient is strictly less than -2^{31} , return -2^{31} .

Input Format

The first line contains an integer `dividend`.

The second line contains an integer `divisor`.

Output Format

Print the quotient after division.

Constraints

$-10^8 \leq \text{dividend}, \text{divisor} \leq 10^8$

`divisor` != 0

Sample Input

DSA Sheet 2 - Problems 21 to 40

10

3

Sample Output

3

Explanation

When dividing 10 by 3: $10 / 3 = 3$ remainder 1. So, the quotient is 3.

--- Problem 40: Subsets (Power Set) ---

Problem Statement

Given an integer array nums of unique elements, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

A subset is a collection of elements selected from the array, including the empty subset and the array itself.

Input Format

The first line contains an integer n - the size of the array.

The second line contains n space-separated integers representing the array nums.

Output Format

Print all possible subsets of the array.

Constraints

$1 \leq \text{nums.length} \leq 10$

$-10 \leq \text{nums}[i] \leq 10$

All the numbers in nums are unique.

Sample Input

3

1 2 3

Sample Output

[[], [1], [2], [1,2], [3], [1,3], [2,3], [1,2,3]]

Explanation

All possible subsets of the array [1,2,3] are:

[[], [1], [2], [3], [1,2], [1,3], [2,3], [1,2,3]]

These form the power set of the given array.